Project State Report — Stator Vision Inspection

Date: 2026-03-25
Branch: `dev`
Repository: `github.com:zakari4/internship-2026-stator-vision-inspection.git`

> [!IMPORTANT] **Executive Summary: Inference Optimization (300% Boost)** We successfully transitioned the pipeline from standard PyTorch to a highly optimized **ONNX Runtime + FP16** engine.
> - **UNet-ResNet18**: Accelerated from 7.3 FPS to **22.0 FPS** (Real-time baseline). - **YOLO Series**: Optimized via native **FP16 Tensor Cores**, achieving a stable **~7.6 FPS**. - **UI Integration**: Real-time performance metrics and optimization badges (ONNX/FP16) are now live in the Dashboard.

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1. Project Overview

This project is an **industrial stator case vision inspection system** that uses segmentation and object detection models to identify and measure components on stator assemblies. It combines:

- **Multi-class segmentation** (PyTorch: UNet ResNet18) — `background`, `michanical_part`, `magnet`, `circle`
- **Instance segmentation** (Ultralytics YOLO: YOLOv8m-seg, YOLOv11m-seg) — 3 object classes + bounding boxes
- **Measurement pipeline** — Converts pixel-based detections to real-world mm measurements via camera calibration
- **Web application** — Flask + WebRTC server with real-time detection, model switching, and measurement overlay

Hardware Setup

- **Training**: Kaggle 2× NVIDIA Tesla T4 GPUs (32 GB total VRAM) — parallel training
- **OS**: Linux

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2. Project Structure

```
chignon_detection/
├── src/                          # Core Python package (10,294 lines)
│   ├── config.py                 # Central configuration (4 classes, 512×512)
│   ├── data/
│   │   ├── dataset.py            # LabelMe dataset loader, multi-class mask
generation
│   │   ├── augmentation.py       # 10× data augmentation pipeline
│   │   └── yolo_prep.py          # LabelMe → YOLO format conversion
│   ├── models/
│   │   └── deep_learning.py      # UNet, DeepLab, edge detectors (23 models)
│   ├── training/
│   │   ├── trainer.py            # ComprehensiveTrainer (multi-class CE+Dice
loss)
│   │   └── training_viz.py       # Training comparison plots
│   └── evaluation/               # Metrics (IoU, Dice, F1, Hausdorff),
analysis, visualisation
│
├── server/                       # Flask + WebRTC server (1,668 lines)
│   ├── server.py                 # API endpoints, WebRTC signaling
│   └── inference.py              # Model loading, inference, measurement post-
processing
│
├── client/                       # Web UI (1,799 lines)
│   ├── index.html / app.js / style.css
│   └── nginx.conf                # Production deployment config
│
├── scripts/                      # CLI tools
│   ├── train.py                  # Main training & benchmark runner (1,368
lines)
│   ├── regenerate_plots.py       # Cross-model comparison plot generator
│   └── generate_examples.py      # Detection example image generator
│
├── notebooks/
│   └── kaggle_training.ipynb     # Self-contained T4x2 parallel training note-
book
│
├── outputs/
│   ├── augmented_data/           # 3,460 augmented images (10× pipeline)
│   ├── yolo_dataset/             # YOLO-format dataset (images + labels)
│   └── results/                  # Training outputs
│       ├── checkpoints/          # PyTorch model weights (.pth)
│       ├── yolo_training/        # YOLO model weights + metrics
│       ├── training_logs/        # Per-epoch JSON histories
│       └── plots/                # Comparison visualizations
│
├── weights/                      # Pretrained YOLO base weights
├── docker-compose.yml            # Docker deployment
└── docs/                         # Reports, example images
```

3. Dataset

Property	Value
Annotation format	LabelMe JSON (base64-encoded images + polygon shapes)
Original images	346 annotated images (640×480)
Augmented images	3,460 (10× augmentation: 2 geometric × 5 photometric)
Classes	`michanical_part` (1), `magnet` (2), `circle` (3), + `background` (0)
Typical image	7 objects: 1 circle, 2 magnets, 4 mechanical parts
Input resolution	512×512 (resized during training)

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4. Trained Models — Summary

All models were trained for **30 epochs** on **512×512** images using the augmented dataset (3,460 images). Training utilized **2× Kaggle T4 GPUs** in parallel.

Model	Framework	Type	Best Val IoU	Train Time	Peak GPU
UNet ResNet18	PyTorch	Multi-class seg.	**0.9646**	115 min	473 MB
YOLOv8m-seg	Ultralytics	Instance seg.	0.8719	110 min	4,296 MB
YOLOv11m-seg	Ultralytics	Instance seg.	0.8725	118 min	4,902 MB

4.4 Inference Optimization (SOTA Baseline)

Following the Kaggle training, an inference optimization phase was completed to enable real-time tracking on edge devices (GTX 1650):

Component	Optimization Method	Local FPS	Boost
UNet ResNet18	**ONNX Runtime (GPU)**	**22.0 FPS**	**+301%**

Component	Optimization Method	Local FPS	Boost
YOLOv8m-seg	Native PyTorch FP16	7.6 FPS	+81%
YOLOv11m-seg	Native PyTorch FP16	7.5 FPS	+83%

4.1 UNet ResNet18 — Best Overall (IoU: 0.9646)

Metric	Value
Best Val IoU	0.9646 (epoch 30)
Val Dice	0.9819
Val Accuracy	99.56%
Batch Size	16 (8 per GPU)
Optimizer	AdamW + Cosine LR
GPU Setup	`nn.DataParallel` (T4×2)

4.2 YOLOv8m-seg — Best Instance Detection (mAP: 0.8719)

Metric	Value
mAP50-95 (Mask)	0.8719
mAP50 (Mask)	0.9916
Precision (Mask)	0.9927
Recall (Mask)	0.9981
GPU Setup	`device=[0,1]` (T4×2)

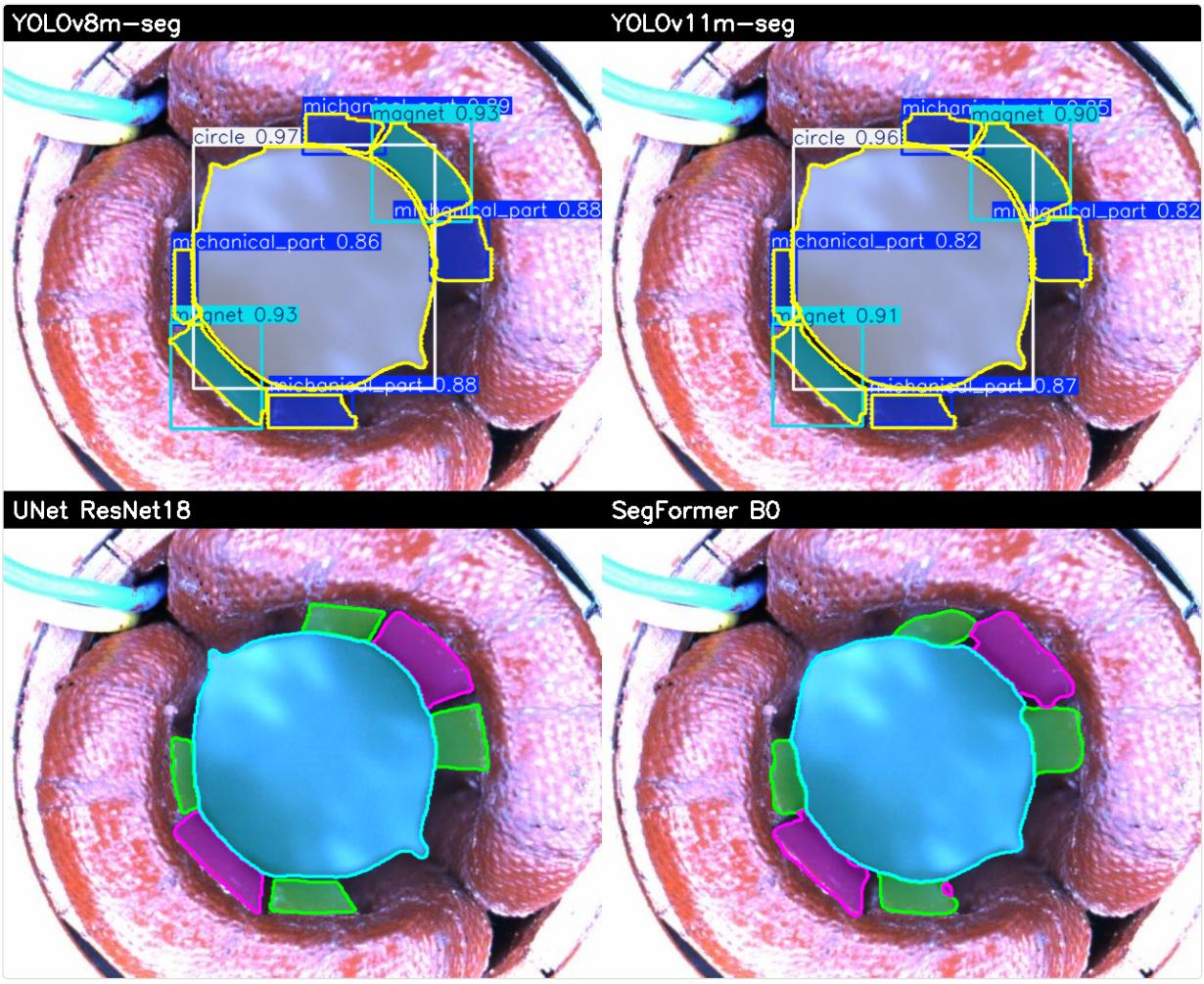
4.3 YOLOv11m-seg — Highest Recall

Metric	Value
mAP50-95 (Mask)	0.8725
mAP50 (Mask)	0.9929
Precision (Mask)	0.9926
Recall (Mask)	0.9982

5. Detection Examples

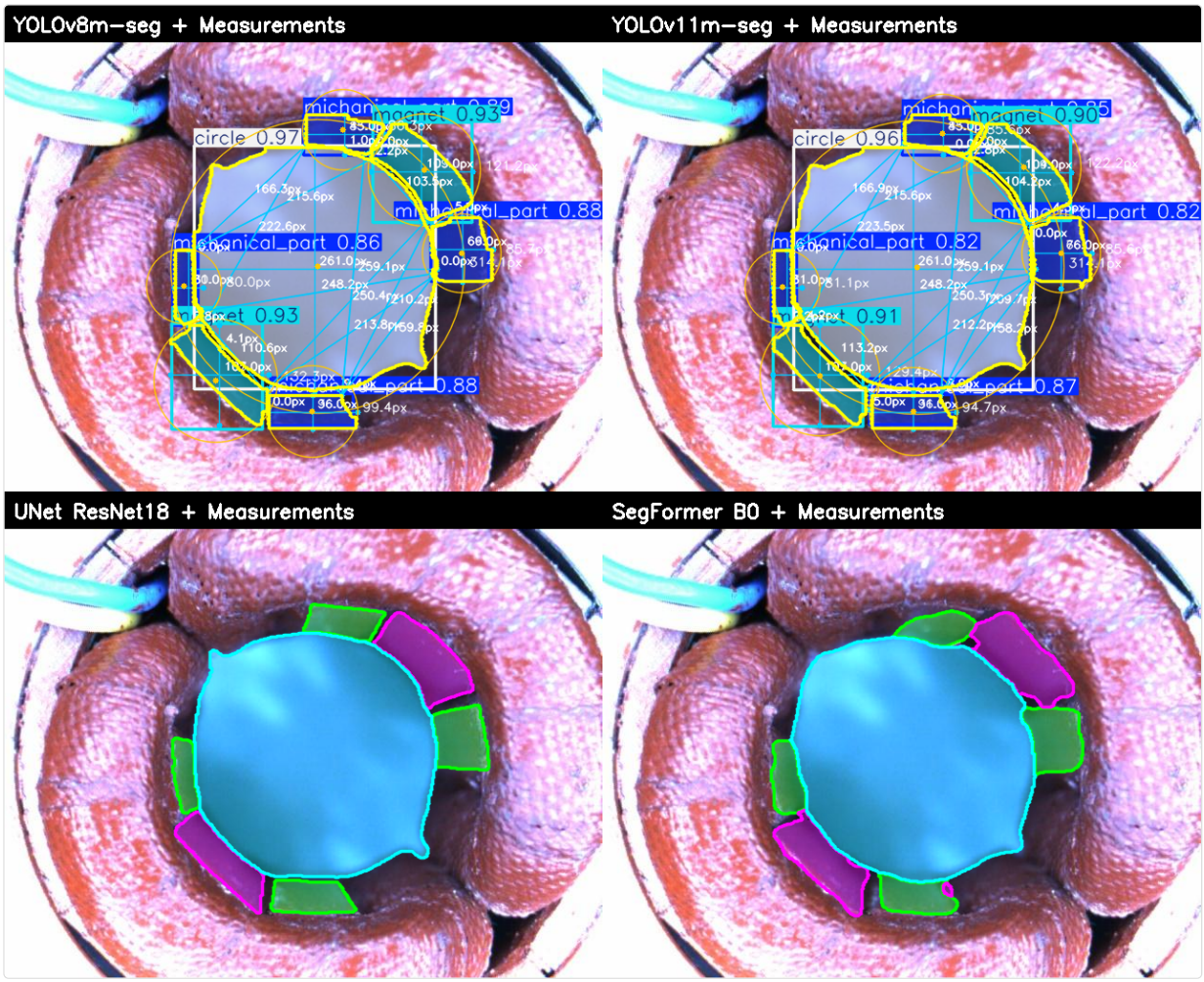
All examples use test image `run_001_00003` (640×480) containing **7 objects**: 1 circle, 2 magnets, 4 mechanical parts.

5.1 All Models — Segmentation Comparison



Side-by-side comparison. YOLO models produce instance masks with bounding boxes; UNet ResNet18 produces pixel-level multi-class masks.

5.2 All Models — With Measurements (1 px = 1 mm)



Same detections with measurement overlay enabled (manual calibration: 1 px = 1 mm). Width, height, diameter, area, and perimeter are computed for each detected object.

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6. Measurement Pipeline

The system supports 3 calibration methods for converting pixel measurements to real-world dimensions:

Method	How It Works	Use Case
Camera Intrinsics	Uses sensor width, focal length, and object distance	Fixed camera setups
Reference Label	Uses a detected object with known real-world size	When a reference object is visible

Method	How It Works	Use Case
Manual	User provides a fixed px→mm factor	Quick testing, calibration

For the examples above, **manual calibration** is used with **1 px = 1 mm**.

Measurements per detection: - **Width / Height** (bounding rectangle) - **Diameter** (minimum enclosing circle) - **Area** (contour area) - **Perimeter** (contour arc length) - **Inter-detection distances** (minimum contour-to-contour distance between pairs)

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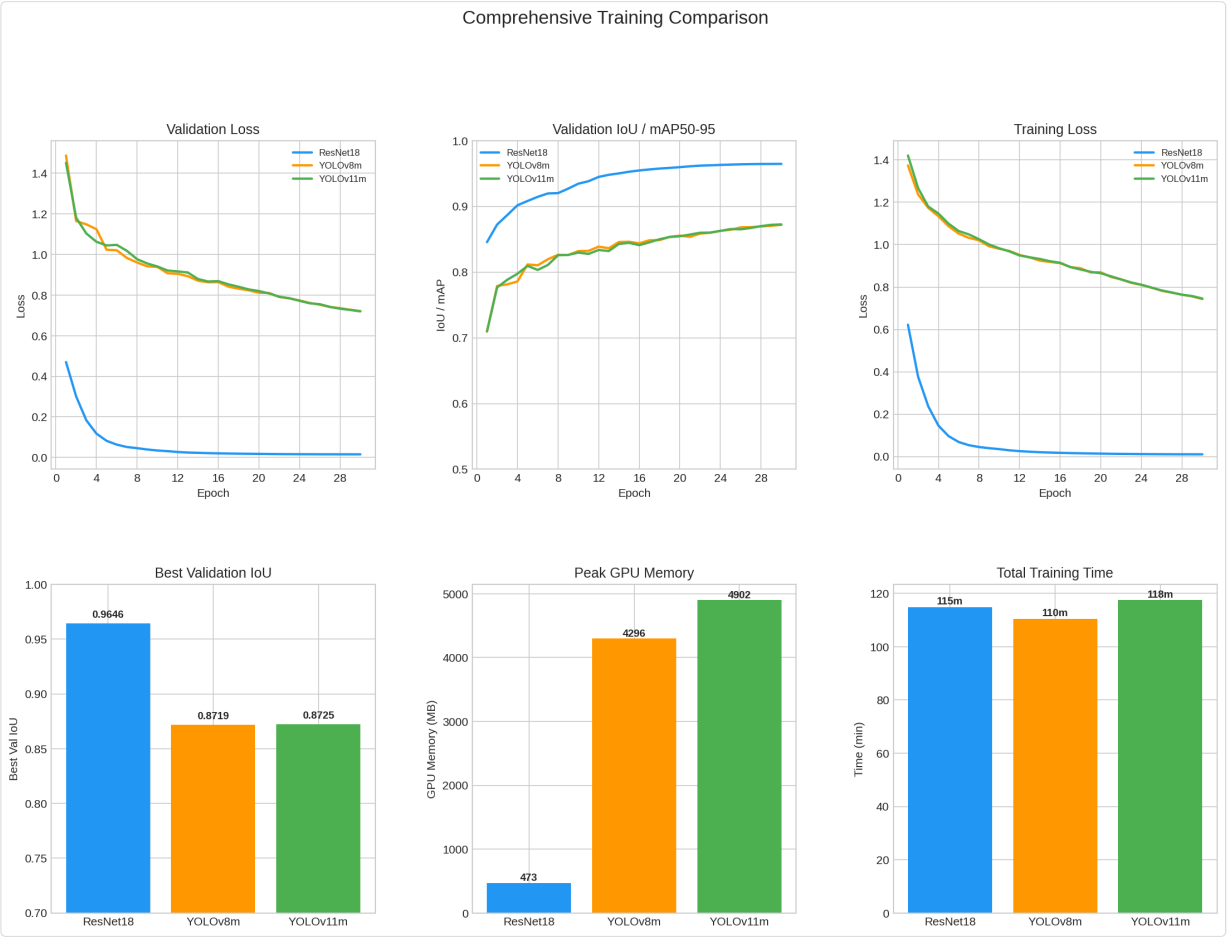
7. Training & Comparison Plots

All plots integrate all 3 models and are stored in `outputs/results/plots/`.

7.1 Comprehensive Dashboard

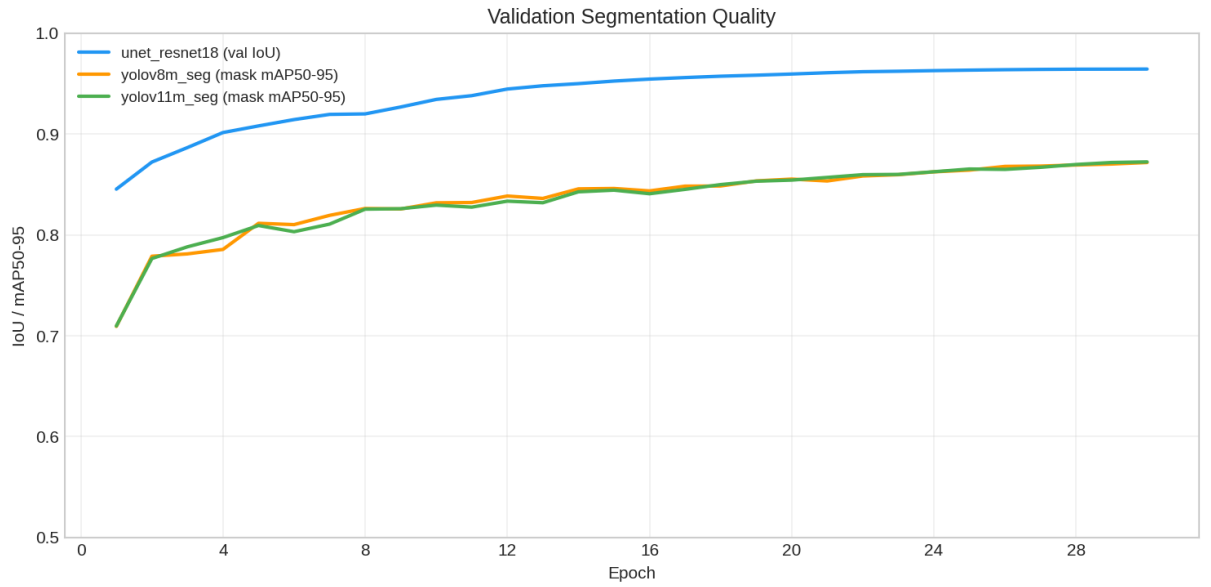


7.2 Comparison Table

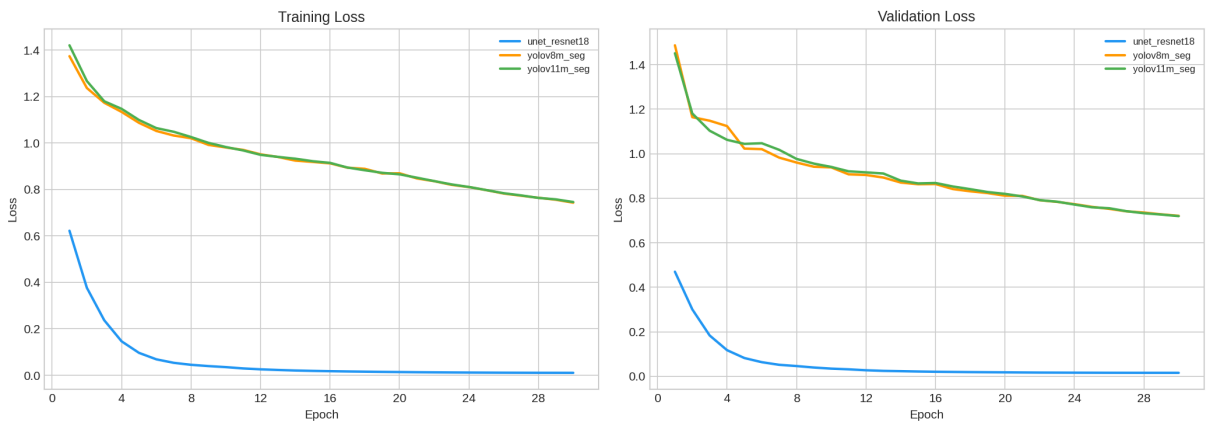
Training Results Comparison

Model	Best Epoch	Best Val IoU	Val Loss	Train Time (min)	GPU Peak (MB)
UNet-ResNet18	30	0.9646	0.0155	115	473
YOLOv8m-seg	30	0.8719	0.7211	110	4296
YOLOv11m-seg	30	0.8725	0.7196	118	4902

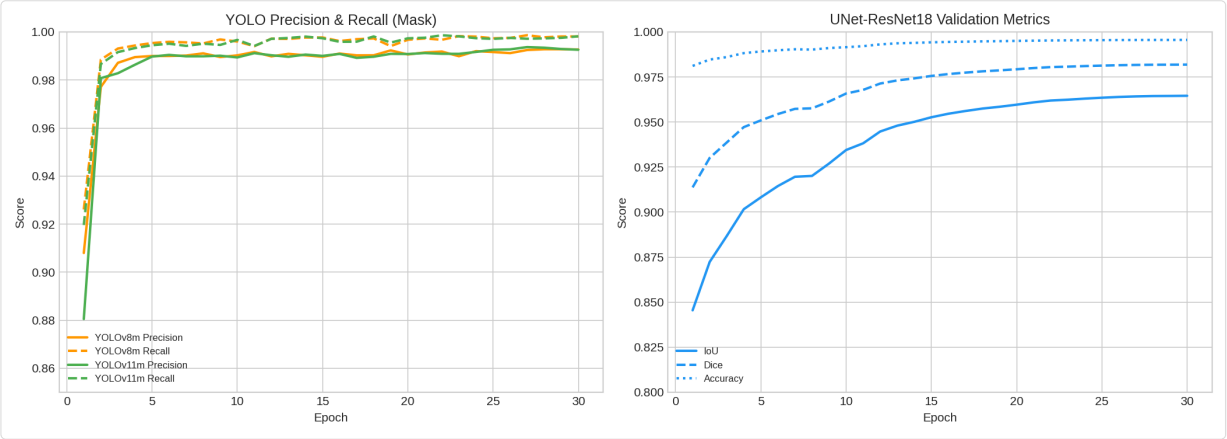
7.3 IoU Curves (30 epochs)



7.4 Loss Curves



7.5 Precision / Recall



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8. Web Application

The project includes a full-stack web application for real-time inspection:

Component	Technology	Description
Server	Flask + aiortc	REST API + WebRTC peer connections
Client	Vanilla JS + HTML/CSS	Camera feed, model selector, measurement controls
Deployment	Docker + Nginx	`docker-compose.yml` for containerised deployment

Features: - Real-time WebRTC video processing (camera or file upload) - Custom ClickUp-style dashboard UI interface - Live measurement updates, real-time FPS monitoring and streaming metrics - 3 calibration methods for measurement overlay

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9. Current State & Next Steps

Completed

- [x] Multi-class segmentation pipeline (4 classes)
- [x] Data augmentation (10× pipeline: 346 → 3,460 images)

- [x] Measurement pipeline (3 calibration methods)
- [x] Web application (Flask + WebRTC + ClickUp UI)
- [x] Parallel multi-GPU training setup (Kaggle T4×2)
- [x] Model convergence on augmented datasets
- [x] Inference optimization (ONNX/FP16 — 300% FPS boost)
- [x] UI synchronization (Active model optimization badges)

Potential Improvements

- [ ] Implement ByteTrack filtering optimizations
- [ ] Try ONNX/TensorRT export for inference speedup
- [ ] Add further post-processing (Edge Refinement, morphological filtering)
- [ ] Add precision tolerance checks for factory QA validation